① $\mathcal{H}^s_\infty(E) = \inf \left\{ \sum_{k=1}^{\infty} (\text{diam}\, U_k)^s : E \subseteq \bigcup_{k=1}^{\infty} U_k \right\}$

To prove : $\dim_H E = \inf\{ s \geq 0 : \mathcal{H}^s_\infty(E) = 0 \}$

Proof : Let $s_0 = \inf\{ s \geq 0 : \mathcal{H}^s_\infty(E) = 0 \}$

For any $\delta > 0$, $\mathcal{H}^s_\infty(E) \leq \mathcal{H}^s_\delta(E) \leq \mathcal{H}^s(E)$

$\therefore \quad \mathcal{H}^s(E) = 0 \Rightarrow \mathcal{H}^s_\infty(E) = 0$

i.e. $\{ s \geq 0 : \mathcal{H}^s(E) = 0 \} \subseteq \{ s \geq 0 : \mathcal{H}^s_\infty(E) = 0 \}$

$\Rightarrow \inf\{ s \geq 0 : \mathcal{H}^s(E) = 0 \} \geq \inf\{ s \geq 0 : \mathcal{H}^s_\infty(E) = 0 \}$

$\Rightarrow \boxed{\dim_H E \geq s_0}$ — (i)

To show : $s_0 \leq \dim_H E$

Suppose $\mathcal{H}^s_\infty(E) = 0$.

Then for $\varepsilon > 0$, $\exists \{U_k\}_{k=1}^{\infty}$ s.t. $E \subseteq \bigcup_{k=1}^{\infty} U_k$

and $\sum_{k=1}^{\infty} (\text{diam}\, U_k)^s < \varepsilon$.

$\Rightarrow (\text{diam}\, U_k)^s < \varepsilon \quad \forall k$

$\Rightarrow \text{diam}\, U_k < \varepsilon^{1/s}$

Choosing $\varepsilon = \delta^s$, we get $\text{diam}\, U_k < \delta \ \forall$

$\therefore \{U_k\}_{k=1}^{\infty}$ is a δ-cover of E.

$\Rightarrow 0 \leq \mathcal{H}^s_\delta(E) < \delta^s$

Taking limit as $\delta \to 0^+$, we get $\mathcal{H}^s(E) = 0$

$\Rightarrow s \geq \dim_H E$

$\therefore \inf\{ s \geq 0 : \mathcal{H}^s_\infty(E) = 0 \} \geq \dim_H E$.

i.e. $\boxed{s_0 \geq \dim_H E}$ — (ii)

(i) & (ii) $\Rightarrow s_0 = \dim_H E$.

② $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

$\dim_H f(S) = \dim_H S$.

$$S = \bigcup_{n=1}^{\infty} \left\{ \left(S \cap [\tfrac{1}{n}, n]\right) \cup S \cap [-n, -\tfrac{1}{n}] \right\} \cup S \cap \{0\}$$

Since f is biLipschitz on $[\frac{1}{n}, n]$ and $[-n, -\frac{1}{n}]$,

$$\dim_H f\left(S \cap [\tfrac{1}{n}, n]\right) = \dim_H \left(S \cap [\tfrac{1}{n}, n]\right)$$

$$\dim_H f\left(S \cap [-n, -\tfrac{1}{n}]\right) = \dim_H \left(S \cap [-n, -\tfrac{1}{n}]\right)$$

$$\dim_H f(S \cap \{0\}) = 0$$

$$\dim_H S = \sup_{n \in \mathbb{N}} \max \left\{ \dim_H S \cap [\tfrac{1}{n}, n], \dim_H S \cap [-n, -\tfrac{1}{n}] \right\}$$

Similarly, for $\dim_H f(S)$.

Let $\varphi_1, \varphi_2, \dots, \varphi_N$ be contraction maps on a complete metric space (X, d) i.e. $\exists\ 0 < c_i < 1$ s.t. $d(\varphi_i(x), \varphi_i(y)) \leq c_i\, d(x,y)$ $\forall x, y \in X$.

for $i = 1, 2, \dots, N$.

We know that $\exists$ a unique nonempty compact set K s.t. $K = \bigcup_{i=1}^{N} \varphi_i(K)$.

(Thus K is called the attractor of the IFS)

<u>Theorem</u>: If $s_0 \geq 0$ is the unique solution of $\sum_{i=1}^{N} c_i^s = 1$, then $d = \dim_H K \leq s_0$.

Also, $\mathcal{H}^{s_0}(K) < \infty$.

Proof: We'll show that $\mathcal{H}^{s_0}(K) = 0$.

Since $K = \bigcup_{i=1}^{N} \varphi_i(K)$,

$$K = \bigcup_{(i_1, i_2, \ldots, i_m)} \varphi_{i_1} \circ \cdots \circ \varphi_{i_m}(K) \quad \text{for any } m \in \mathbb{N}.$$

Now, $\operatorname{diam}\left(\varphi_{i_1} \circ \cdots \circ \varphi_{i_m}(K)\right) \leq c_{i_1} \cdots c_{i_m} \operatorname{diam}(K)$
$\leq c^m \operatorname{diam}(K)$,

where $c = \max\{c_1, c_2, \ldots, c_N\}$

For any $\delta > 0$, if we choose m s.t. $c^m \operatorname{diam}(K) < \delta$,
then $\{\varphi_{i_1} \circ \cdots \circ \varphi_{i_m}(K) : 1 \leq i_1, \ldots, i_m \leq N\}$ forms a δ-cover of K

$$\therefore \quad \mathcal{H}^s_\delta(K) \leq \sum_{i_1, \ldots, i_m = 1}^{N} \left(\operatorname{diam} \varphi_{i_1} \circ \cdots \circ \varphi_{i_m}(K)\right)^s$$
$$\leq \sum_{i_1, \ldots, i_m} (c_{i_1} \cdots c_{i_m})^s (\operatorname{diam} K)^s$$

Since $\sum_{i=1}^{N} c_i^{s_0} = 1$, $\sum_{i_1, \ldots, i_m = 1}^{N} (c_{i_1} \cdots c_{i_m})^{s_0} = 1$

$$\therefore \quad \mathcal{H}^{s_0}_\delta(K) \leq (\operatorname{diam} K)^{s_0}$$

$$\Rightarrow \quad \mathcal{H}^{s_0}(K) \leq (\operatorname{diam} K)^{s_0} < \infty$$

$$\Rightarrow \quad \dim_H(K) \leq s_0 \quad \left(\begin{array}{l}\text{because if } s_0 < \dim_H(K), \\ \text{then } \mathcal{H}^{s_0}(K) = \infty\end{array}\right)$$

Consider the IFS on $\mathbb{R}^n$ with similarity maps $S_1, S_2, \ldots, S_N$, where similarity ratios $c_i < 1$.

$$\|S_i(x) - S_i(y)\| = c_i \|x - y\| \quad \forall x, y.$$

Similarity dimension of K is defined as $s_0 \geq 0$ s.t. $\sum_{i=1}^{N} c_i^{s_0} = 1$

By the previous theorem, we know that

$$\dim_H K \le s_0 .$$

Also, $\mathcal{H}^{s_0}(K) < \infty$

Suppose the IFS satisfies the "open set condition":

$\exists$ a nonempty open set V s.t.

$$S_i(V) \subseteq V \quad \forall i = 1, 2, \cdots, N$$

and $S_i(V) \cap S_j(V) = \phi$ for $i \neq j$.

Then one can prove that $\boxed{\dim_H K = s_0}$

Also, $0 < \mathcal{H}^{s_0}(K) < \infty$.